

AMALIY MASHG'ULOT-1

MAVZU: Axborotlarning kompyuterda tasvirlanishi.

Ishning maqsadi:

- a) Talabalarga axborot tushunchasi, ularning turlari axborotni to'plash, saqlash, qayta ishlash va uzatish tushunchalarini etkazish;
- b) Talabalarga axborotlarni mashina xotirasida tasvirlanishini o'rgatish.
- c) Axborot o'lchov birliklari haqida ma'lumot berish.

Nazariy ma'lumotlar:

Axborotlarning o'lchov birliklari ichida eng kichigi bit. Bir bit axborot-bu bitta ikkilik raqam: 0 yoki 1. Bir bit birlik juda kichik axborot bo'lagi bo'lgani uchun, amalda ko'proq bayt birligi ishlatiladi. Bir bayt 8 bitdan iborat. 1 baytda 256 tagacha simvollarni va xarflarni kodlash mumkin. Masalan, ulardan 52 tasi ingliz tili alfaviti uchun, 66 tasi rus alfaviti uchun, 10 tasi raqamlar uchun va qolgan 128 tasi boshqa simvollar uchun mo'ljallangan.

Bir bayt nafaqat axborot birligi, balki EHMlar xotirasining elementar yacheykasi hamdir. EHM xotirasi shunday yacheykalarning ketma ketligidan tashkil topgan. Xotira hajmini o'lchashda baytdan kattaroq birliklardan kilobayt (Kbayt), Megobayt (Mbayt), Gigobayt (Gbt) Terabayt (Tbt) lardan ko'p foydalaniladi.

1 bayt = 8 bit

1 Kbt = 1024 bayt = 2^{10} bt

1 Mbt = 1024 Kbt = 1048576 bayt = 2^{20} bt

1 Gbt = 1024 Mbt = 1073741824 bayt = 2^{30} bt

1 Tbt = 1024 Gbt = 1 099 511 627 776 bayt = 2^{40} bt

EHM bu elektron-raqamli qurilma. Elektron qurilma deyilishiga sabab, u har qanday axborotni elektron signallar yordamida qayta ishlaydi. Raqamli qurilma deyilishiga sabab, har qanday axborot sonlar ko'rinishida beriladi.

Masalaning qo'yilishi:

1-vazifa:

Kitob sahifalari soni $512+G+N$ ta, har bir sahifada $50+N$ ta satr bor, har bir satrda $60+N$ ta belgi bor, bitta belgi 1 bayt, yani 8 bit. Bu kitobning hajmi qancha?

2-vazifa:

Kitobni internetdan yuklab olish uchun qancha vaqt ketadi. Modening tezligi 64 Kbit/sekund, 128 Kbit/sekund, 256 Kbit/sekund, 512 Kbit/sekund.

3-vazifa:

1.44 Mbayt hajmli floppi, 720 Mbayt hajmli CD, 4.7 Gbayt hajmli DVD, 1 Gbayt flesh card, 2 Gbayt flesh card, 4 Gbayt flesh card, 8 Gbayt flesh card, xotira qurilmalariga nechtdan **kitob** joy bo'ladi.

Eslatma: G- talabaning guruh nomeri, N talabaning guruh jurnalidagi nomeri;

Topshiriqni bajarish namunasi.

1-vazifa:

Kitob sahifalari soni 512 ta, har bir sahifada 50 ta satr bor, har bir satrda 60 ta belgi bor, bitta belgi 1 bayt, yani 8 bit. Bu kitobning hajmi qancha?

Sahifalar=512;

Qatorlar=50;

Belgilar=60;

Hajm= Sahifalar* Qatorlar* Belgilar*8 bit=512*50*60*8=1228800 bit

2-vazifa:

Kitobni internetdan yuklab olish uchun qancha vaqt ketadi. Modening tezligi 128 Kbit/sekund.

Hajm=1228800 bit

Tezlik=128 Kbit/sekund=128*1024 bit=131072 bit

Vaqt=Hajm/Tezlik=1228800 bit /131072 bit =9,375 sekund

3-vazifa:4.7 Gbayt hajmli DVD diskga nechta **kitob** joy bo'ladi.

Hajm_DVD=4,7*1073741824 bayt=4,7*1073741824*8=40372692582 bit

Hajm_kitob=1228800 bit;

Kitolar_soni= Hajm_DVD/ Hajm_kitob=40372692582 bit/1228800 bit=32855 ta

Sanoq sistemalari ustida amallar bajarish.

Topshiriqni bajarish namunasi.

1) 110010111 berilgan ikkilik sanoq sistemasidagi sonni o'nlik sanoq sistemasiga o'kazish.

$$1 * 2^8 + 1 * 2^7 + 0 * 2^6 + 0 * 2^5 + 1 * 2^4 + 0 * 2^3 + 1 * 2^2 + 1 * 2^1 + 1 * 2^0 = 407_{10}$$

berilgan ikkilik son o'nlik sanoq sistemasida 407 soniga tengekan.

2) ED7₁₆ O'n oltilik sanoq sistemasidagi sonni o'nlik sanoq sistemasiga otkazish.

$$E = 14, D = 13; \quad 14 * 16^2 + 13 * 16^1 + 7 * 16^0 = 3799_{10}$$

3) 3652₈ Sakkizlik sanoq sistemasida berilgan sonni o'nlik sanoq sistemasiga otkazish.

$$3 * 8^3 + 6 * 8^2 + 5 * 8^1 + 2 * 8^0 = 1962_{10}$$

4) 100110₂ va 110₂ sonlarning arifmetik yig'indisini toping. Bu sonlarni bir ustunga yozib, umumiy qoida bo'yicha qo'shamiz.

100110₂

+ 110₂

101100₂

5) 1011,101₂ va 101,01₂ sonlarining ayirmasini toping.

1011,101₂

- 101,01₂

110,011₂

6) 1010₂ va 11₂ sonlarning ko'paytmasini toping.

1010₂

x 11₂

1010

1010

11110₂

7) 732₈

8) 732₈

+ 324₈

- 324₈

1256₈ 406₈

9) 437₁₀ sonini ikkilik sistemada yozing:

Son	Bo'luvchi	Qoldiq
437	2	1
218	2	0
109	2	1
54	2	0
27	2	1
13	2	1
6	2	0
3	2	1
1	2	1

Agar qoldiqlarni teskari tartibda yozib chiqsak, kerakli natija hosil bo'ladi:

$$437_{10} = 110110101_2.$$

10) 7465_{10} sonini sakkizlik sistemada ifodalang:

Son	Bo'luvchi	qoldiq
7465	8	1
933	8	5
116	8	4
14	8	6
1	8	0

$$\text{Natija: } 7465_{10} = 6451_8$$

11) 98653_{10} sonini 16-lik sistemada ifodalang:

Son	Bo'luvchi	qoldiq
98653	16	13
6165	16	5
385	16	1
24	16	8
1	16	0

$$\text{Natija : } 98653_{10} = 815D_{16}$$

12) 25_{10} ni ikkilik sistemaga o'tkazing:

25	2				
24	12	2			
1	12	6	2		
	0	6	3	2	
		0	2	1	
			1		$25_{10} = 11001_2$

13) 25_{10} ni sakkizlik sistemaga o'tkazing:

25	8				
24	3				
1					$25_{10} = 31_8$

14) 28_{10} ni o'n oltilik sistemaga o'tkazing:

28	16				
16	1				
12					$28_{10} = 1C_{16}$

15) $0,3125_{10}$ sonini ikkilik sanoq sistemasiga o'tkazing.

0	3125	$0,3125_{10} = 0,0101_2$
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	6250	
0	250	
1	50	
0	0	
1		

16) $0,12_{10}$ sonini ikkilik sanoq sistemasiga o'tkazing.

0	12	$0,12_{10} = 0,000111_2$
0	24	
0	48	
0	96	
1	92	
1	84	
1	68	
1		

1-amaliy mashg'ulot topshiriqlari.

Har bir talaba 1- topshirig'ni bajarishdan oldin sanoq sistemaliri, har xil sanoq sistemalarida bajariladigan arifmetik amallar, sonlarni bir sanoq sistemasidan boshqa sanoq sistemasiga o'tkazish bo'yicha amaliy mashg'ulotning nazariy qismini yoritadi. Oxirida mashg'ulotlarni bajaradi:

1-vazifa:

$398 + G + N_{10}$ berilgan o'nlik sanoq sistemasidagi sonni X_2, X_8 va X_{16} sanoq sistemalariga o'tkazing. Buyerda G – talabaning guruh nomeri, N – talabaning guruh jurnalidagi nomeri.

2-vazifa:

Birinchi vazifadan hosil bo'lgan X_2, X_8 va X_{16} sonlarni o'nlik sanoq sistemasiga o'tkazing.

3-vazifa:

Birinchi vazifadan hosil bo'lgan X_2 ikkilik sanoq sistemasidagi sonni X_2, X_8 va X_{16} lik sanoq sistemalariga **triada** va **tetrada** usulida o'tkazing.

4-vazifa:

Talabaning guruh jurnalidagi nomeriga mos variantni tanlaydi va berilgan amallarni ketma – ket bajaradi.

1 - variant

- 1) $1000101_2 \rightarrow X_{10}$ 2) $124_{10} \rightarrow X_2$ 3) $FF_{16} \rightarrow X_{10}$
 4) $224_8 * 346_8 = X_{16}$ 5) $2DF_{16} * 237_8 = X_{10}$ 6) $1101001_2 * 5FD_{16} = X_8$
 7) $567_8 - 456_8 = X_{16}$ 8) $10111001_2 - 100111_2 = X_8$ 9) $FDA_{16} - BDE_{16} = X_{10}$
 10) $675_8 + 345_8 = X_{10}$ 11) $65,46_8 * 47,53_8 = X_{16}$ 12) $6F, AC_{16} * 3B, FE_{16} = X_8$ hisoblang.

2 - variant

- 1) $110101011_2 \rightarrow X_{10}$ 2) $153_{10} \rightarrow X_2$ 3) $A254_{16} \rightarrow X_{10}$
 4) $524_8 * 376_8 = X_{16}$ 5) $6DF_{16} * 237_8 = X_{10}$ 6) $1101001_2 * 8FD_{16} = X_8$
 7) $667_8 - 456_8 = X_{16}$ 8) $10111001_2 - 110111_2 = X_8$ 9) $FDA_{16} - BCE_{16} = X_{10}$
 10) $645_8 + 345_8 = X_{10}$ 11) $67,46_8 * 47,54_8 = X_{16}$ 12) $6F, AC_{16} * 3B, FA_{16} = X_8$ hisoblang.

3 - variant

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|-------------------------------------|----------------------------------|--|
| 1) $110111010_2 \rightarrow X_{10}$ | 2) $537_{10} \rightarrow X_2$ | 3) $3ABC_{16} \rightarrow X_{10}$ |
| 4) $324_8 * 345_8 = X_{16}$ | 5) $2DF_{16} * 236_8 = X_{10}$ | 6) $1101001_2 * 5FB_{16} = X_8$ |
| 7) $567_8 - 457_8 = X_{16}$ | 8) $10111001_2 - 100101_2 = X_8$ | 9) $FDA_{16} - BDC_{16} = X_{10}$ |
| 10) $675_8 + 346_8 = X_{10}$ | 11) $65,46_8 * 47,57_8 = X_{16}$ | 12) $6F, AC_{16} * 3B, CE_{16} = X_8$ hisoblang. |

4 - variant

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|-------------------------------------|----------------------------------|--|
| 1) $111111011_2 \rightarrow X_{10}$ | 2) $6734_{10} \rightarrow X_2$ | 3) $8901_{16} \rightarrow X_{10}$ |
| 4) $354_8 * 345_8 = X_{16}$ | 5) $2DF_{16} * 226_8 = X_{10}$ | 6) $1101001_2 * 5EB_{16} = X_8$ |
| 7) $567_8 - 447_8 = X_{16}$ | 8) $10111001_2 - 10101_2 = X_8$ | 9) $FDA_{16} - BDA_{16} = X_{10}$ |
| 10) $675_8 + 343_8 = X_{10}$ | 11) $65,46_8 * 47,52_8 = X_{16}$ | 12) $6F, AC_{16} * 3B, FE_{16} = X_8$ hisoblang. |

5 - variant

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|-------------------------------------|----------------------------------|--|
| 1) $110101110_2 \rightarrow X_{10}$ | 2) $7894_{10} \rightarrow X_2$ | 3) $A2B_{16} \rightarrow X_{10}$ |
| 4) $554_8 * 345_8 = X_{16}$ | 5) $2DF_{16} * 226_8 = X_{10}$ | 6) $1101001_2 * 5EB_{16} = X_8$ |
| 7) $667_8 - 247_8 = X_{16}$ | 8) $10111011_2 - 101011_2 = X_8$ | 9) $FDA_{16} - CDA_{16} = X_{10}$ |
| 10) $675_8 + 143_8 = X_{10}$ | 11) $67,46_8 * 42,52_8 = X_{16}$ | 12) $6F, A8_{16} * 3B, FE_{16} = X_8$ hisoblang. |

6 - variant

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|------------------------------------|----------------------------------|--|
| 1) $11011011_2 \rightarrow X_{10}$ | 2) $4873_{10} \rightarrow X_2$ | 3) $1ABE_{16} \rightarrow X_8$ |
| 4) $524_8 * 345_8 = X_{16}$ | 5) $5DF_{16} * 226_8 = X_{10}$ | 6) $110111_2 * 5EB_{16} = X_8$ |
| 7) $637_8 - 241_8 = X_{16}$ | 8) $10111010_2 - 10111_2 = X_8$ | 9) $FDA_{16} - CD2_{16} = X_{10}$ |
| 10) $675_8 + 142_8 = X_{10}$ | 11) $67,56_8 * 42,52_8 = X_{16}$ | 12) $5F, F8_{16} * 3B, 2E_{16} = X_8$ hisoblang. |

7 - variant

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|-------------------------------------|----------------------------------|--|
| 1) $110111111_2 \rightarrow X_{10}$ | 2) $3784_{10} \rightarrow X_2$ | 3) $ABF2_{16} \rightarrow X_{10}$ |
| 4) $324_8 * 345_8 = X_{16}$ | 5) $5DF_{16} * 626_8 = X_{10}$ | 6) $110101_2 * 5EB_{16} = X_8$ |
| 7) $632_8 - 241_8 = X_{16}$ | 8) $1011101_2 - 10111_2 = X_8$ | 9) $FDA_{16} - CB2_{16} = X_{10}$ |
| 10) $625_8 + 144_8 = X_{10}$ | 11) $62,56_8 * 41,52_8 = X_{16}$ | 12) $5F, F8_{16} * 3B, BE_{16} = X_8$ hisoblang. |

8 - variant

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|--------------------------------------|----------------------------------|--|
| 1) $1101101101_2 \rightarrow X_{10}$ | 2) $3518_{10} \rightarrow X_2$ | 3) $AB41_{16} \rightarrow X_8$ |
| 4) $124_8 * 645_8 = X_{16}$ | 5) $5DF_{16} * 666_8 = X_{10}$ | 6) $110101_2 * 4EB_{16} = X_8$ |
| 7) $632_8 - 246_8 = X_{16}$ | 8) $10111001_2 - 10111_2 = X_8$ | 9) $FDA_{16} - 5B2_{16} = X_{10}$ |
| 10) $655_8 + 144_8 = X_{10}$ | 11) $66,56_8 * 41,55_8 = X_{16}$ | 12) $5F, FB_{16} * 3B, BE_{16} = X_8$ hisoblang. |

9 - variant

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|-------------------------------------|----------------------------------|--|
| 1) $101101101_2 \rightarrow X_{10}$ | 2) $1345_{10} \rightarrow X_2$ | 3) $A95_{16} \rightarrow X_8$ |
| 4) $524_8 * 645_8 = X_{16}$ | 5) $6DF_{16} * 666_8 = X_{10}$ | 6) $11010111_2 * 4EB_{16} = X_8$ |
| 7) $632_8 - 346_8 = X_{16}$ | 8) $101111101_2 - 10111_2 = X_8$ | 9) $FDA_{16} - 5B8_{16} = X_{10}$ |
| 10) $655_8 + 744_8 = X_{10}$ | 11) $67,56_8 * 46,55_8 = X_{16}$ | 12) $6F, FB_{16} * 7B, BE_{16} = X_8$ hisoblang. |

10 - variant

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|-------------------------------------|-----------------------------------|--|
| 1) $111011101_2 \rightarrow X_{10}$ | 2) $5432_{10} \rightarrow X_2$ | 3) $34567_8 \rightarrow X_{16}$ |
| 4) $564_8 * 645_8 = X_{16}$ | 5) $9DF_{16} * 666_8 = X_{10}$ | 6) $11010111_2 * 8EB_{16} = X_8$ |
| 7) $637_8 - 346_8 = X_{16}$ | 8) $101111101_2 - 101111_2 = X_8$ | 9) $FDA_{16} - 8B8_{16} = X_{10}$ |
| 10) $635_8 + 744_8 = X_{10}$ | 11) $67,76_8 * 46,75_8 = X_{16}$ | 12) $8F, FB_{16} * 7B, BE_{16} = X_8$ hisoblang. |

11 - variant

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|------------------------------------|--------------------------------|----------------------------------|
| 1) $11010111_2 \rightarrow X_{10}$ | 2) $9542_{10} \rightarrow X_2$ | 3) $4537_8 \rightarrow X_{16}$ |
| 4) $564_8 * 645_8 = X_{16}$ | 5) $9DF_{16} * 666_8 = X_{10}$ | 6) $11010111_2 * 8EB_{16} = X_8$ |

- 7) $637_8-346_8=X_{16}$ 8) $101111101_2-101111_2=X_8$ 9) $FDA_{16}-8B_{16}=X_{10}$
 10) $635_8+744_8=X_{10}$ 11) $67,76_8*46,75_8=X_{16}$ 12) $8F,FB_{16}*7B,BE_{16}=X_8$ hisoblang.

12 - variant

- 1) $11110010_2 \rightarrow X_{10}$ 2) $2323_8 \rightarrow X_2$ 3) $2BC_{16} \rightarrow X_8$
 4) $577_8*642_8=X_{16}$ 5) $99F_{16}*126_8=X_{10}$ 6) $11010111_2*A2B_{16}=X_8$
 7) $637_8-346_8=X_{16}$ 8) $10101111_2-1011_2=X_8$ 9) $FDA_{16}-CB2_{16}=X_{10}$
 10) $325_8+745_8=X_{10}$ 11) $45,76_8*46,75_8=X_{16}$ 12) $8F,FB_{16}*8B,BE_{16}=X_8$ hisoblang.

13 - variant

- 1) $100001101_2 \rightarrow X_{10}$ 2) $7762_{10} \rightarrow X_2$ 3) $345_8 \rightarrow X_{16}$
 4) $523_8*642_8=X_{16}$ 5) $9DF_{16}*726_8=X_{10}$ 6) $11010111_2*A8B_{16}=X_8$
 7) $437_8-346_8=X_{16}$ 8) $10111101_2-1111_2=X_8$ 9) $FDA_{16}-7BC_{16}=X_{10}$
 10) $725_8+744_8=X_{10}$ 11) $43,76_8*76,75_8=X_{16}$ 12) $8F,4B_{16}*2B,BE_{16}=X_8$ hisoblang.

14 - variant

- 1) $111101111_2 \rightarrow X_{10}$ 2) $8927_{10} \rightarrow X_8$ 3) $5432_8 \rightarrow X_{16}$
 4) $571_8*641_8=X_{16}$ 5) $9DC_{16}*126_8=X_{10}$ 6) $110111_2*A5B_{16}=X_8$
 7) $437_8-346_8=X_{16}$ 8) $11011101_2-1111_2=X_8$ 9) $FDA_{16}-CB5_{16}=X_{10}$
 10) $725_8+744_8=X_{10}$ 11) $53,76_8*56,75_8=X_{16}$ 12) $8F,7B_{16}*7B,BE_{16}=X_8$ hisoblang.

15 - variant

- 1) $10110111_2 \rightarrow X_8$ 2) $4567_{10} \rightarrow X_2$ 3) $AB2F_{16} \rightarrow X_{10}$
 4) $544_8*647_8=X_{16}$ 5) $7DC_{16}*526_8=X_{10}$ 6) $110111_2*AB_{16}=X_8$
 7) $427_8-356_8=X_{16}$ 8) $1111101_2-1111_2=X_8$ 9) $FDA_{16}-CB5_{16}=X_{10}$
 10) $765_8+724_8=X_{10}$ 11) $57,76_8*56,76_8=X_{16}$ 12) $8F,4B_{16}*7B,BE_{16}=X_8$ hisoblang.

16 - variant

- 1) $10100110_2 \rightarrow X_8$ 2) $7654_8 \rightarrow X_2$ 3) $5674_{10} \rightarrow X_{16}$
 4) $514_8*627_8=X_{16}$ 5) $7DC_{16}*536_8=X_{10}$ 6) $11111_2*AB_{16}=X_8$
 7) $467_8-256_8=X_{16}$ 8) $1111101_2-1111_2=X_8$ 9) $FDA_{16}-CB5_{16}=X_{10}$
 10) $715_8+724_8=X_{10}$ 11) $51,76_8*56,76_8=X_{16}$ 12) $F,4B_{16}*7B,BE_{16}=X_8$ hisoblang.

17 - variant

- 1) $110101011_2 \rightarrow X_8$ 2) $24578_{10} \rightarrow X_{16}$ 3) $10101101_2 \rightarrow X_{10}$
 4) $534_8*627_8=X_{16}$ 5) $7D2_{16}*546_8=X_{10}$ 6) $111011_2*AB_{16}=X_8$
 7) $567_8-256_8=X_{16}$ 8) $111111_2-1111_2=X_8$ 9) $FDA_{16}-2B5_{16}=X_{10}$
 10) $765_8+734_8=X_{10}$ 11) $51,36_8*56,26_8=X_{16}$ 12) $F,3B_{16}*7B,BE_{16}=X_8$ hisoblang.

18 - variant

- 1) $100110101_2 \rightarrow X_{10}$ 2) $89765_{10} \rightarrow X_2$ 3) $3745_8 \rightarrow X_{16}$
 4) $564_8*227_8=X_{16}$ 5) $7D2_{16}*246_8=X_{10}$ 6) $111011_2*DB_{16}=X_8$
 7) $561_8-251_8=X_{16}$ 8) $111111_2-1101_2=X_8$ 9) $FD6_{16}-2B5_{16}=X_{10}$
 10) $365_8+734_8=X_{10}$ 11) $52,36_8*56,26_8=X_{16}$ 12) $F,4B_{16}*7B,BE_{16}=X_8$ hisoblang.

19 - variant

- 1) $101010101_2 \rightarrow X_8$ 2) $98983_{10} \rightarrow X_{16}$ 3) $5732_8 \rightarrow X_{10}$
 4) $764_8*223_8=X_{16}$ 5) $7D2_{16}*746_8=X_{10}$ 6) $1110011_2*DB_{16}=X_8$
 7) $564_8-251_8=X_{16}$ 8) $11001111_2-1101_2=X_8$ 9) $FD3_{16}-2B5_{16}=X_{10}$
 10) $465_8+634_8=X_{10}$ 11) $55,36_8*52,26_8=X_{16}$ 12) $F,2B_{16}*B,BE_{16}=X_8$ hisoblang.

20 - variant

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|-------------------------------------|----------------------------------|---|
| 1) $110110011_2 \rightarrow X_{10}$ | 2) $37891_{10} \rightarrow X_8$ | 3) $7356_8 \rightarrow X_{16}$ |
| 4) $464_8 * 243_8 = X_{16}$ | 5) $7D6_{16} * 446_8 = X_{10}$ | 6) $10110011_2 * DB_{16} = X_8$ |
| 7) $562_8 - 451_8 = X_{16}$ | 8) $1101111_2 - 10101_2 = X_8$ | 9) $FD3_{16} - DB5_{16} = X_{10}$ |
| 10) $415_8 + 634_8 = X_{10}$ | 11) $55,36_8 * 57,26_8 = X_{16}$ | 12) $F,2B_{16} * A, BE_{16} = X_8$ hisoblang. |

21 - variant

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|-------------------------------------|----------------------------------|---|
| 1) $100111011_2 \rightarrow X_{16}$ | 2) $98906_{10} \rightarrow X_8$ | 3) $ABC_{16} \rightarrow X_2$ |
| 4) $264_8 * 243_8 = X_{16}$ | 5) $5D6_{16} * 446_8 = X_{10}$ | 6) $1110011_2 * DB_{16} = X_8$ |
| 7) $662_8 - 451_8 = X_{16}$ | 8) $1111111_2 - 10101_2 = X_8$ | 9) $FD4_{16} - DBF_{16} = X_{10}$ |
| 10) $445_8 + 234_8 = X_{10}$ | 11) $54,36_8 * 54,26_8 = X_{16}$ | 12) $F3,2B_{16} * A3, BE_{16} = X_8$ hisoblang. |

22 - variant

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|-----------------------------------|----------------------------------|--|
| 1) $1000101_2 \rightarrow X_{10}$ | 2) $124_{10} \rightarrow X_2$ | 3) $FF1_{16} \rightarrow X_{10}$ |
| 4) $224_8 * 243_8 = X_{16}$ | 5) $2D6_{16} * 446_8 = X_{10}$ | 6) $10110011_2 * DB_{16} = X_8$ |
| 7) $612_8 - 451_8 = X_{16}$ | 8) $11111011_2 - 10101_2 = X_8$ | 9) $FD6_{16} - DBF_{16} = X_{10}$ |
| 10) $665_8 + 234_8 = X_{10}$ | 11) $64,36_8 * 54,26_8 = X_{16}$ | 12) $F3,B_{16} * A3, BE_{16} = X_8$ hisoblang. |

23 - variant

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|-------------------------------------|----------------------------------|--|
| 1) $111000111_2 \rightarrow X_{10}$ | 2) $87245_{10} \rightarrow X_8$ | 3) $37013_8 \rightarrow X_{16}$ |
| 4) $324_8 * 243_8 = X_{16}$ | 5) $4D6_{16} * 146_8 = X_{10}$ | 6) $10110011_2 * CB_{16} = X_8$ |
| 7) $642_8 - 451_8 = X_{16}$ | 8) $11101011_2 - 10101_2 = X_8$ | 9) $FB6_{16} - DBF_{16} = X_{10}$ |
| 10) $445_8 + 234_8 = X_{10}$ | 11) $24,36_8 * 54,26_8 = X_{16}$ | 12) $F7,B_{16} * A3, BE_{16} = X_8$ hisoblang. |

24 - variant

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|-------------------------------------|------------------------------------|--|
| 1) $101010101_2 \rightarrow X_{10}$ | 2) $87934_{10} \rightarrow X_{16}$ | 3) $ABC_{16} \rightarrow X_2$ |
| 4) $524_8 * 443_8 = X_{16}$ | 5) $4D6_{16} * 746_8 = X_{10}$ | 6) $100110011_2 * CB_{16} = X_8$ |
| 7) $643_8 - 451_8 = X_{16}$ | 8) $11101011_2 - 11101_2 = X_8$ | 9) $FBD_{16} - DBF_{16} = X_{10}$ |
| 10) $465_8 + 234_8 = X_{10}$ | 11) $54,36_8 * 53,26_8 = X_{16}$ | 12) $F6,B_{16} * A3, BE_{16} = X_8$ hisoblang. |

25 - variant

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|--------------------------------------|----------------------------------|--|
| 1) $1101011011_2 \rightarrow X_{10}$ | 2) $3672_8 \rightarrow X_{16}$ | 3) $139A_{16} \rightarrow X_2$ |
| 4) $534_8 * 473_8 = X_{16}$ | 5) $7D6_{16} * 646_8 = X_{10}$ | 6) $110011_2 * CB_{16} = X_8$ |
| 7) $673_8 - 251_8 = X_{16}$ | 8) $11101011_2 - 1101_2 = X_8$ | 9) $FAD_{16} - DBF_{16} = X_{10}$ |
| 10) $475_8 + 234_8 = X_{10}$ | 11) $42,36_8 * 53,26_8 = X_{16}$ | 12) $F7,B_{16} * A6, BE_{16} = X_8$ hisoblang. |